

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

View Only

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q3 E-Commerce Order Processing – Built-in Exceptions

10
marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 ArrayIndexOutOfBoundsException handled

Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 InputMismatchException handled Invalid input

Your Code

```
1 import java.util.*;  
2 class Order{  
3     public static void main(String[] args)
```

```
4      Scanner sc = new Scanner(System.in);
5      String[] products = {"Laptop", "phone", "watch"};
6      double[] prices = {50000, 20000, 2000};
7      try{
8          int index=sc.nextInt();
9          int quantity=sc.nextInt();
10         if(quantity<=0)
11             throw new IllegalArgumentException("Invalid quantity");
12         double total=prices[index]*quantity;
13         System.out.printf("Total cost: %.2f\n", total);
14     }catch (ArrayIndexOutOfBoundsException e){
15         System.out.println("ArrayIndexOutOfBoundsException");
16     }
17     }catch (InputMismatchException e){
18         System.out.println("InputMismatchException");
19     }catch (IllegalArgumentException e){
20         System.out.println("Invalid quantity");
21     }
22     System.out.println("Transaction completed");
23 }
24 }
25 }
```



Input

1 2

Output

Total cost: 40000.000000
Transaction completed

✓ Submitted